

Model Description

Tactical asset allocation model results from Jan 1997 to Jun 2026 are based on dual momentum model holding the best performing asset. Absolute momentum based trend following filter is used to switch any selected assets that have a negative excess return over the risk free rate to Vanguard Inter-Term Treasury Inv (VFITX). The model uses multiple weighted performance windows to rank the assets by weighted average of asset performance. Tactical asset allocation model trades are executed using the end of month close price each month based on the end of month signals. The time period was constrained by the available data for Vanguard Total Intl Stock Index Inv (VGTSX) [May 1996 - Jun 2026].

Tactical Asset Allocation Model Assets

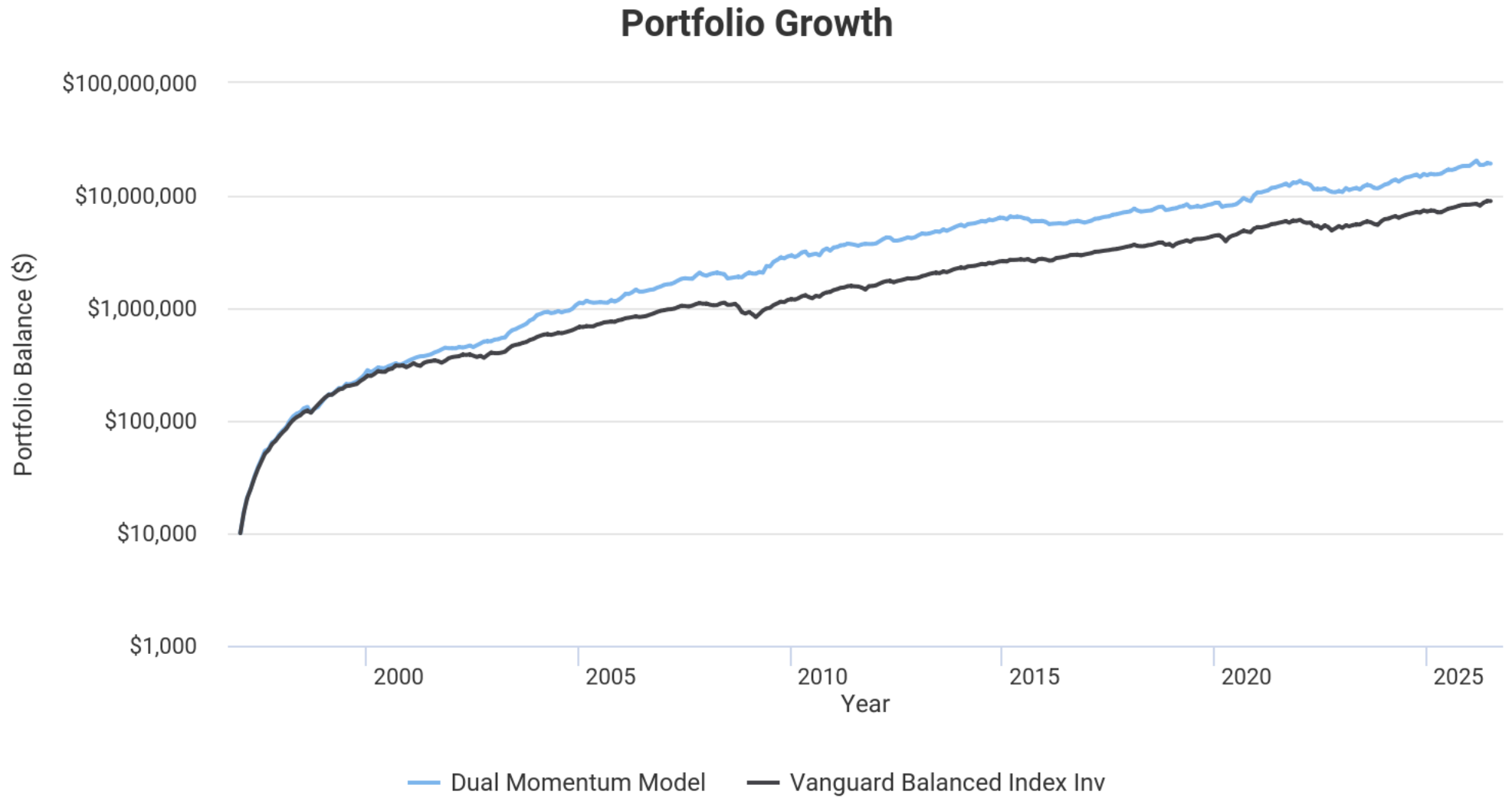
Ticker	Name
VGTSX	Vanguard Total Intl Stock Index Inv
VFINX	Vanguard 500 Index Investor

Lookback Periods

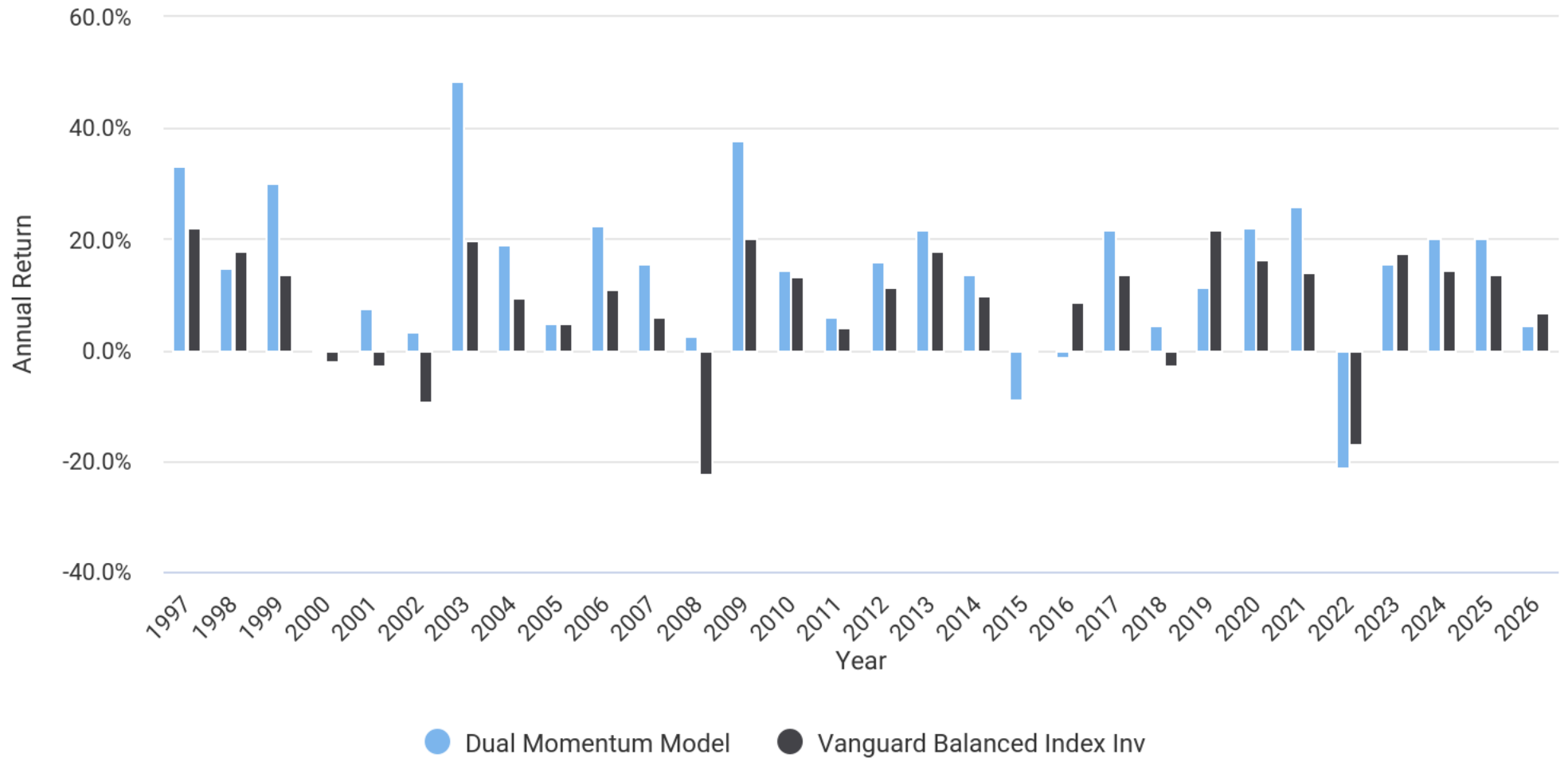
Period	Length	Unit	Weight
Lookback Period 1	1	Months	33.00%
Lookback Period 2	3	Months	33.00%
Lookback Period 3	6	Months	34.00%

Portfolio Performance (Jan 1997 - Jun 2026)

Metric	Dual Momentum Model	Vanguard Balanced Index Inv
Start Balance	\$10,000	\$10,000
End Balance	\$19,276,026	\$8,945,861
Annualized Return (CAGR)	29.23%	25.91%
Time-Weighted Rate of Return (TWRR)	13.52%	7.93%
Money-Weighted Rate of Return (MWRR)	12.44%	8.18%
Standard Deviation	12.14%	9.82%
Best Year	48.66%	22.24%
Worst Year	-21.40%	-22.21%
Maximum Drawdown	-21.04%	-25.62%
Max. Drawdown (excluding cashflows)	-21.66%	-32.57%
Sharpe Ratio	0.93	0.60
Sortino Ratio	1.54	0.88
Active Return	5.59%	N/A
Tracking Error	9.63%	N/A
Information Ratio	0.58	N/A
Benchmark Correlation	0.63	N/A



Annual Returns



Trailing Returns

Name	Total Return			Annualized Return				Annualized Standard Deviation	
	3 Month	Year To Date	1 year	3 year	5 year	10 year	Full	3 year	5 year
Dual Momentum Model	2.77%	4.54%	12.34%	15.96%	8.84%	11.89%	13.52%	11.68%	13.05%
Vanguard Balanced Index Inv	9.57%	6.92%	15.16%	13.89%	7.37%	9.62%	7.93%	9.48%	11.44%

Trailing return and volatility are as of last calendar month ending June 2026

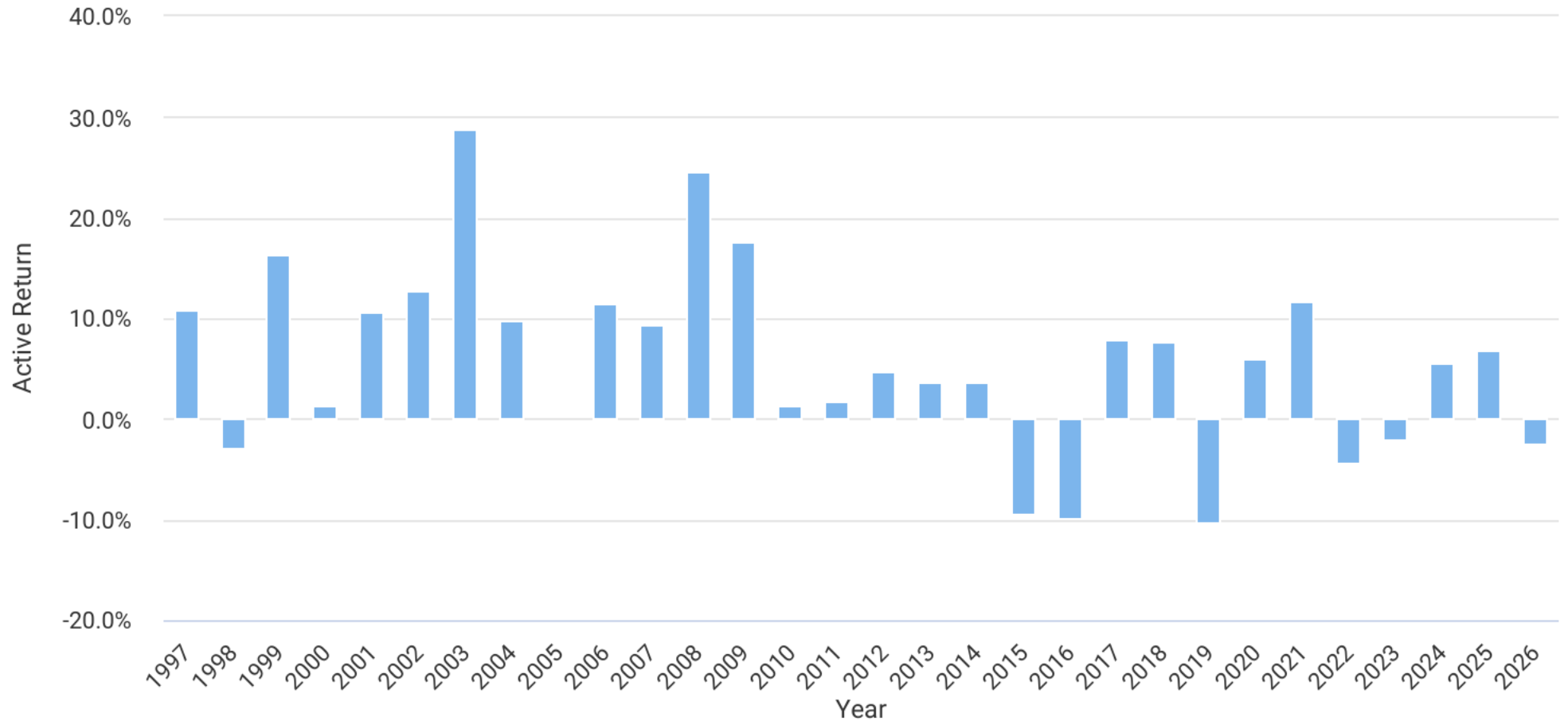
Risk and Return Metrics (Jan 1997 - Jun 2026)

Metric	Dual Momentum Model	Vanguard Balanced Index Inv
Arithmetic Mean (monthly)	1.12%	0.68%
Arithmetic Mean (annualized)	14.34%	8.45%
Geometric Mean (monthly)	1.06%	0.64%
Geometric Mean (annualized)	13.52%	7.93%
Standard Deviation (monthly)	3.51%	2.84%
Standard Deviation (annualized)	12.14%	9.82%
Downside Deviation (monthly)	2.02%	1.83%
Maximum Drawdown	-21.66%	-32.57%
Benchmark Correlation	0.63	1.00
Beta (*)	0.78	1.00
Alpha (annualized)	7.10%	-0.00%
R Squared	40.20%	100.00%
Sharpe Ratio	0.93	0.60
Sortino Ratio	1.54	0.88
Treynor Ratio (%)	14.34	5.90
Calmar Ratio	1.85	1.88
Modigliani–Modigliani Measure	11.34%	8.13%
Active Return	5.59%	N/A
Tracking Error	9.63%	N/A
Information Ratio	0.58	N/A
Skewness	-0.23	-0.60
Excess Kurtosis	1.69	1.18
Historical Value-at-Risk (5%)	4.60%	4.41%
Analytical Value-at-Risk (5%)	4.64%	3.99%
Conditional Value-at-Risk (5%)	6.88%	6.16%
Upside Capture Ratio (%)	117.61	100.00
Downside Capture Ratio (%)	78.90	100.00
Positive Periods	241 out of 354 (68.08%)	233 out of 354 (65.82%)
Gain/Loss Ratio	1.08	0.94

(*) Vanguard Balanced Index Inv is used as the benchmark for calculations. Value-at-risk metrics are monthly values.

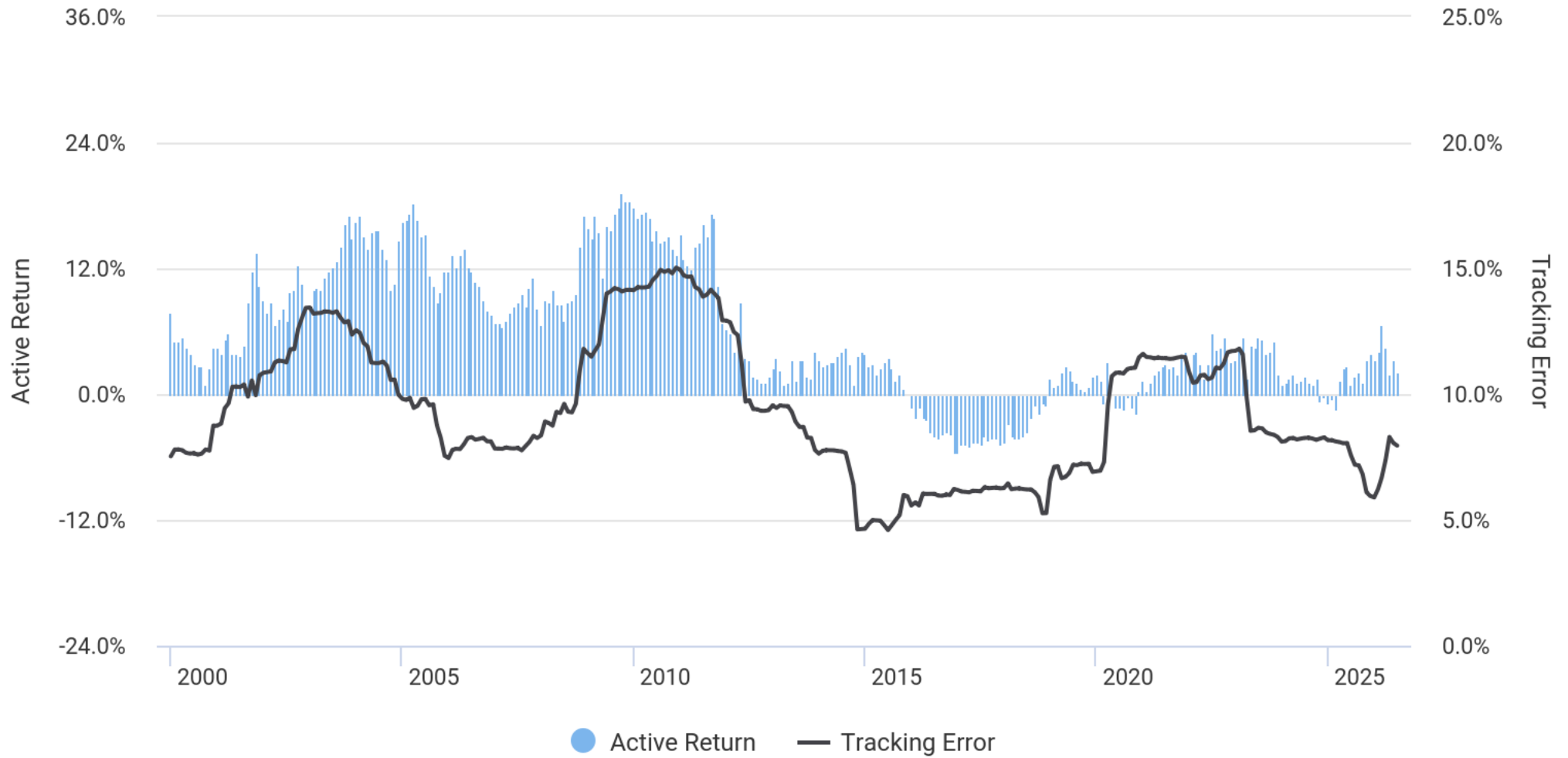
Annualized Active Return

Dual Momentum Model vs. Vanguard Balanced Index Inv



Rolling Active Return and Risk (36 months)

Dual Momentum Model vs. Vanguard Balanced Index Inv



Regime Performance

Regime	Months #	Months %	Portfolio						Benchmark			Risk-free Return	Up/Down Capture			
			Return	Stdev	SR	XS	TE	IR	Return	Stdev	SR		Win %	Upside	Downside	Capture Ratio
High Volatility	102	28.90%	6.87%	14.24%	0.37	9.07%	14.07%	0.64	-2.20%	13.90%	-0.26	2.34%	59.80%	73.52	42.43	1.73
Medium Volatility	176	49.86%	11.41%	12.18%	0.78	1.84%	7.43%	0.25	9.57%	8.08%	0.91	2.16%	57.95%	130.78	131.94	0.99
Low Volatility	75	21.25%	29.19%	7.57%	3.20	9.95%	6.07%	1.64	19.24%	4.86%	3.27	2.36%	72.00%	151.71	127.10	1.19

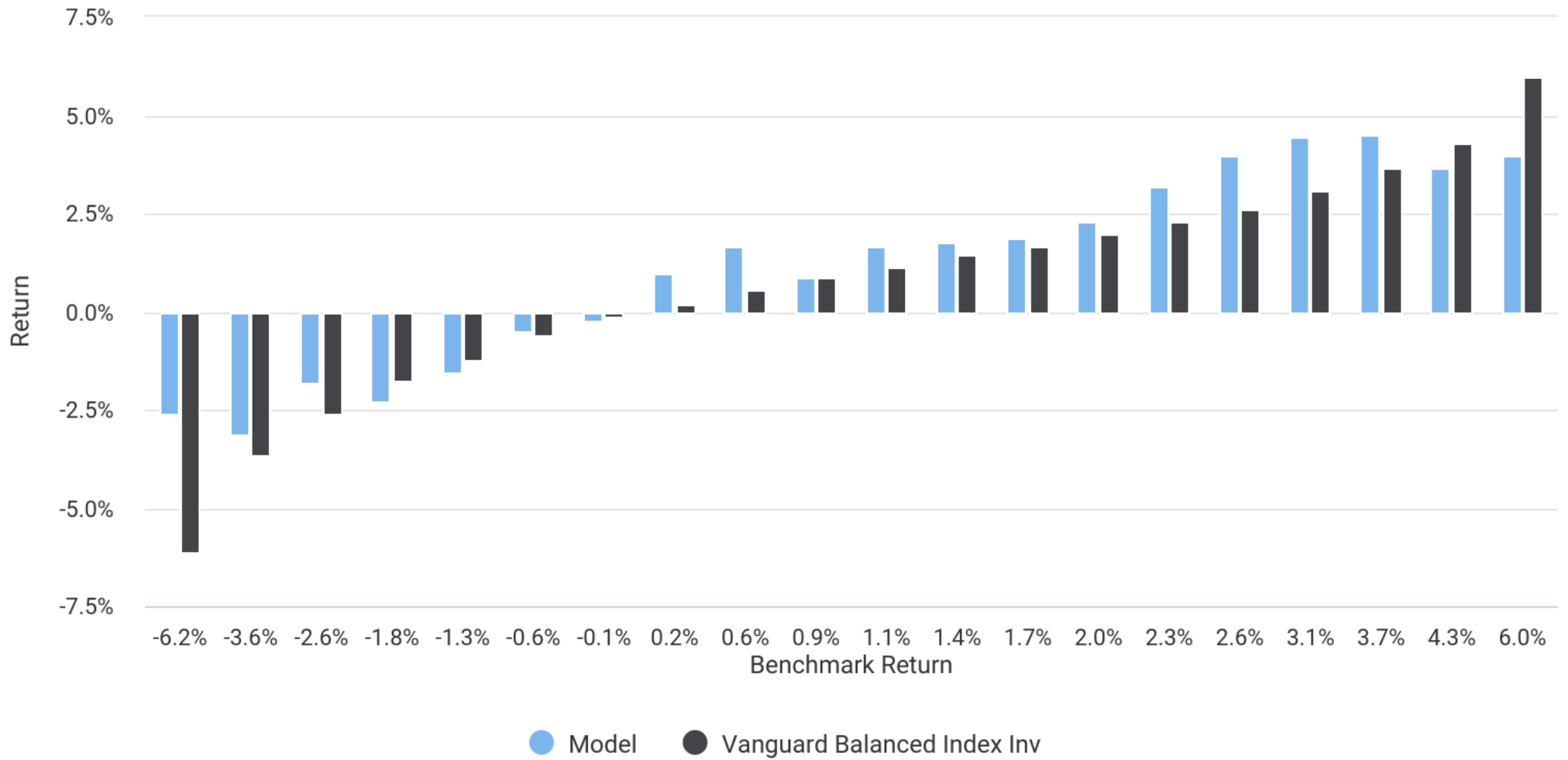
Regime periods based on market regime definitions of Market Volatility

Up vs. Down Market Performance - Dual Momentum Model vs. Vanguard Balanced Index Inv

Market Type	Occurrences					Average Active Return		
	Above Benchmark	Below Benchmark	Total	% Above Benchmark		Above Benchmark	Below Benchmark	Total
Up Market	160	74	234	68%		1.63%	-2.36%	0.37%
Down Market	57	63	120	48%		3.10%	-1.67%	0.59%
Total	217	137	354	61%		2.02%	-2.04%	0.45%

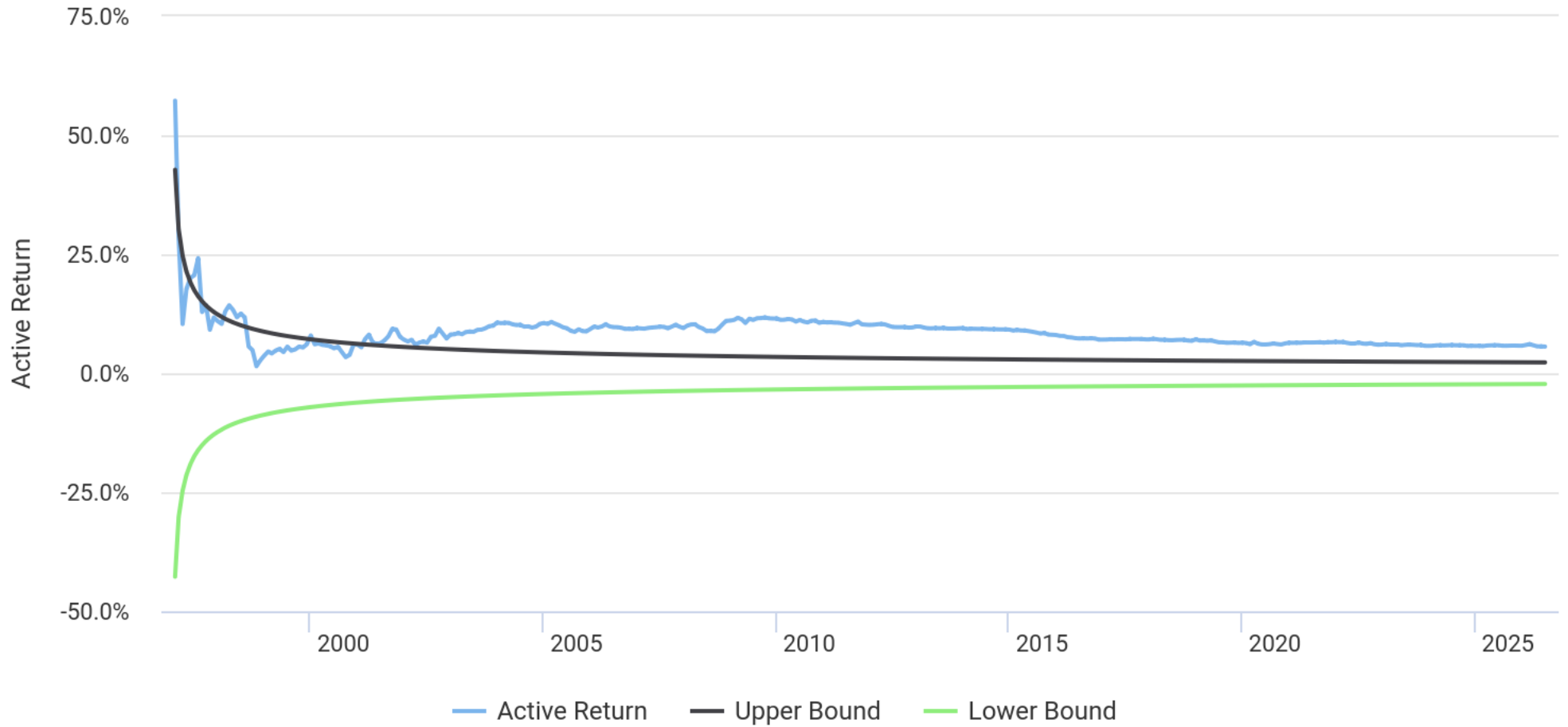
Return vs. Benchmark

Dual Momentum Model vs. Vanguard Balanced Index Inv



Manager Skill (80% confidence)

Dual Momentum Model vs. Vanguard Balanced Index Inv



Dual Momentum Model Returns

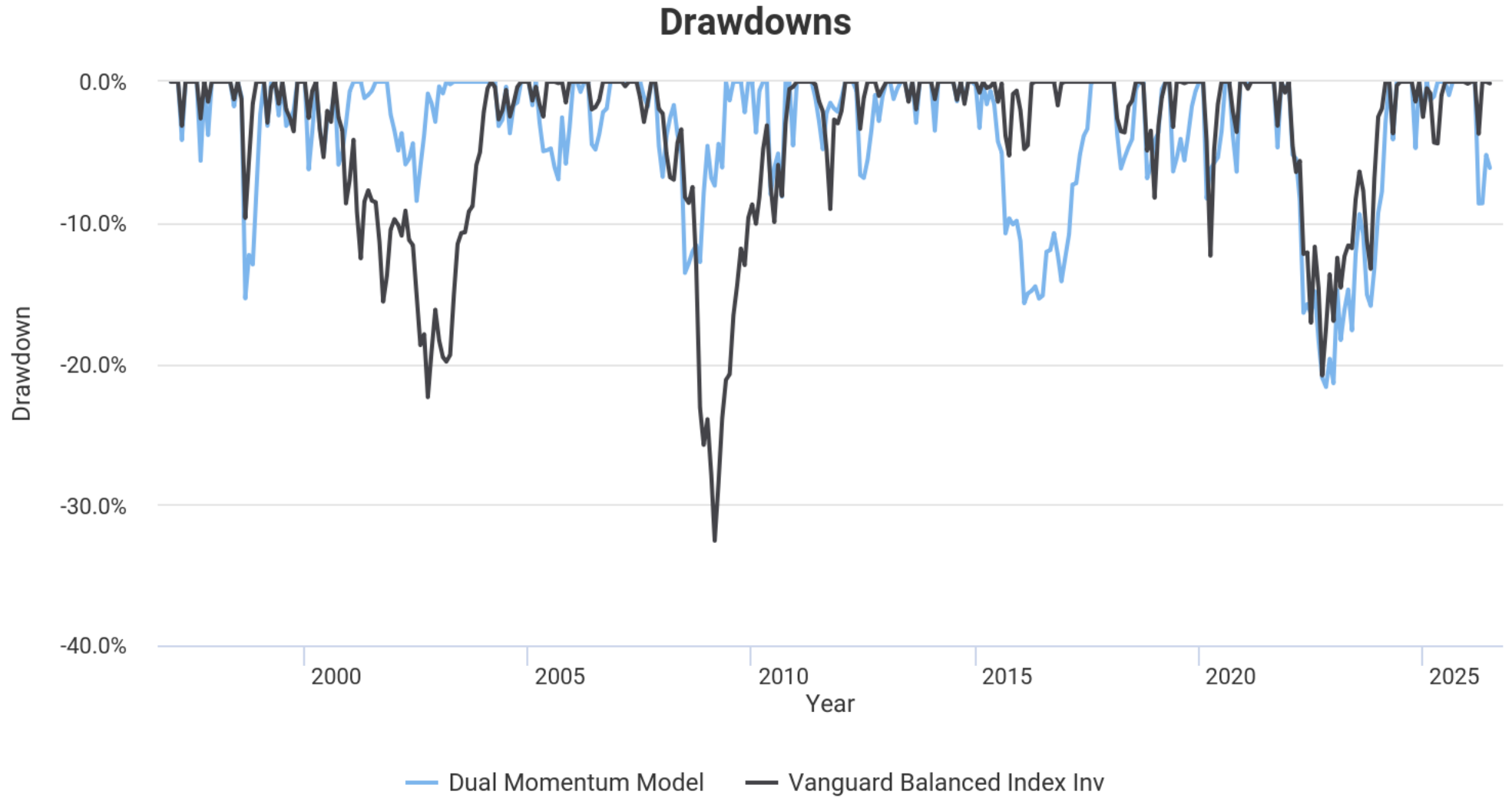
Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Total	Inflation	Cashflow	Balance
1997	6.22%	0.79%	-4.14%	5.96%	6.09%	4.45%	7.97%	-5.61%	5.47%	-3.35%	4.60%	1.72%	33.19%	1.70%	\$60,725	\$81,604
1998	1.11%	7.19%	5.10%	1.01%	-1.74%	4.07%	-1.05%	-14.47%	3.64%	-0.77%	6.07%	5.81%	14.92%	1.61%	\$61,668	\$158,704
1999	4.20%	-3.12%	4.00%	3.85%	-2.39%	5.56%	-3.13%	0.33%	0.82%	3.74%	4.08%	9.37%	29.98%	2.68%	\$63,017	\$280,519
2000	-6.22%	2.83%	3.48%	-3.00%	-2.04%	2.49%	0.62%	1.62%	-5.29%	0.90%	2.14%	2.39%	-0.68%	3.39%	\$65,145	\$345,504
2001	1.33%	1.26%	0.50%	-1.16%	0.22%	0.30%	2.73%	1.11%	2.41%	2.22%	-2.32%	-1.16%	7.55%	1.55%	\$66,986	\$440,142
2002	-1.47%	1.29%	-2.30%	0.42%	1.16%	-4.28%	2.66%	2.42%	3.03%	-0.70%	-1.32%	2.58%	3.23%	2.38%	\$68,049	\$525,063
2003	-0.49%	1.77%	-0.19%	8.25%	5.26%	1.26%	3.18%	2.51%	2.78%	6.49%	2.13%	7.80%	48.66%	1.88%	\$69,593	\$868,385
2004	1.60%	2.50%	0.45%	-3.14%	0.46%	2.40%	-3.31%	2.07%	0.19%	3.24%	7.23%	4.44%	19.16%	3.26%	\$71,456	\$1,115,797
2005	-1.67%	4.68%	-2.93%	-2.07%	0.08%	0.13%	-1.41%	-0.91%	4.73%	-3.37%	2.67%	5.14%	4.65%	3.42%	\$73,881	\$1,245,807
2006	6.87%	-0.72%	3.17%	5.12%	-4.45%	-0.38%	1.15%	1.62%	0.25%	3.25%	1.89%	3.09%	22.40%	2.54%	\$76,264	\$1,607,947
2007	0.96%	0.06%	2.80%	3.92%	3.09%	0.61%	-1.06%	-0.72%	6.38%	5.90%	-4.57%	-2.28%	15.52%	4.08%	\$78,440	\$1,939,402
2008	3.10%	1.35%	0.93%	-2.14%	-1.32%	-8.99%	0.77%	0.98%	0.51%	-1.36%	5.58%	3.66%	2.35%	0.09%	\$81,451	\$2,069,504
2009	-2.39%	-0.58%	3.20%	-1.76%	14.35%	-1.32%	9.96%	3.50%	5.15%	-2.17%	3.86%	1.95%	37.63%	2.72%	\$81,162	\$2,945,475
2010	-3.60%	3.09%	6.01%	1.58%	-8.01%	2.07%	1.07%	-3.22%	10.41%	3.55%	-4.52%	6.67%	14.45%	1.50%	\$82,493	\$3,462,129
2011	0.70%	3.42%	0.03%	2.95%	-1.15%	-1.67%	-2.05%	2.70%	0.73%	-0.42%	-0.23%	1.00%	6.00%	2.96%	\$85,097	\$3,756,008
2012	4.46%	4.31%	3.28%	-0.64%	-6.02%	-0.22%	1.37%	2.24%	2.58%	-1.86%	1.82%	4.17%	16.04%	1.74%	\$86,858	\$4,450,749
2013	3.34%	-1.23%	0.77%	1.91%	2.33%	-1.35%	5.07%	-2.91%	3.12%	3.41%	3.03%	2.51%	21.57%	1.50%	\$88,130	\$5,509,012
2014	-3.47%	4.56%	0.82%	0.72%	2.33%	2.05%	-1.39%	3.98%	-1.41%	2.42%	2.68%	-0.26%	13.51%	0.76%	\$89,560	\$6,349,057
2015	-3.02%	5.74%	-1.59%	0.95%	-0.94%	-2.69%	-0.80%	-6.05%	1.19%	-0.47%	0.29%	-1.59%	-9.04%	0.73%	\$89,666	\$5,859,630
2016	-4.98%	0.82%	0.22%	0.37%	-1.02%	0.25%	3.68%	0.13%	1.37%	-1.72%	-2.15%	1.96%	-1.33%	2.07%	\$90,797	\$5,873,942
2017	1.88%	3.96%	0.10%	2.20%	1.39%	0.52%	3.42%	0.64%	1.81%	1.94%	0.78%	1.10%	21.55%	2.11%	\$92,731	\$7,240,194
2018	5.71%	-3.69%	-2.56%	0.83%	0.76%	0.61%	3.71%	3.25%	0.55%	-6.85%	0.96%	1.90%	4.62%	1.91%	\$94,996	\$7,671,116
2019	0.58%	3.20%	0.72%	4.04%	-6.36%	1.02%	1.43%	-1.59%	1.86%	2.15%	1.11%	3.01%	11.30%	2.29%	\$96,718	\$8,640,651
2020	-0.05%	-8.24%	2.27%	0.46%	0.42%	1.98%	5.63%	7.18%	-3.81%	-2.67%	12.93%	5.82%	22.22%	1.36%	\$97,911	\$10,678,685
2021	-0.10%	2.32%	1.69%	5.32%	0.69%	2.31%	2.38%	3.03%	-4.66%	6.99%	-0.71%	4.47%	25.86%	7.04%	\$102,511	\$13,555,001
2022	-5.19%	-0.28%	-3.12%	-8.73%	0.74%	-0.83%	1.92%	-4.09%	-3.26%	-0.83%	2.55%	-2.15%	-21.40%	6.45%	\$110,714	\$10,758,246
2023	8.40%	-4.15%	2.64%	1.69%	-3.38%	6.60%	3.20%	-1.60%	-4.78%	-0.93%	3.17%	4.53%	15.38%	3.35%	\$115,272	\$12,534,678
2024	1.67%	5.33%	3.21%	-4.10%	4.95%	3.58%	1.20%	2.41%	2.12%	-4.71%	5.86%	-2.40%	20.06%	2.89%	\$118,672	\$15,174,618
2025	2.77%	-1.32%	0.27%	1.23%	4.62%	3.89%	-0.95%	2.02%	3.62%	2.33%	0.23%	0.05%	20.24%	2.68%	\$121,884	\$18,378,508
2026	5.74%	5.31%	-8.64%	0.02%	3.75%	-0.96%							4.54%	3.05%	\$62,558	\$19,276,026

Annual return for 2026 is from 01/01/2026 to 06/30/2026. The cashflow column shows the total annual portfolio contributions (positive) and withdrawals (negative).

Vanguard Balanced Index Inv Returns

Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Total	Inflation	Cashflow	Balance
1997	3.38%	0.00%	-3.12%	3.34%	4.57%	3.16%	5.70%	-2.61%	4.01%	-1.42%	2.19%	1.51%	22.24%	1.70%	\$60,725	\$78,829
1998	0.80%	4.32%	3.16%	0.91%	-1.24%	2.46%	-1.24%	-8.53%	4.53%	4.25%	3.96%	3.96%	17.85%	1.61%	\$61,668	\$160,263
1999	2.49%	-2.90%	2.59%	2.89%	-1.56%	2.93%	-1.97%	-0.63%	-0.96%	3.89%	2.08%	4.33%	13.61%	2.68%	\$63,017	\$250,390
2000	-2.57%	1.98%	3.97%	-3.33%	-2.10%	3.47%	-0.79%	4.90%	-2.53%	-0.94%	-5.38%	1.84%	-2.04%	3.39%	\$65,145	\$308,986
2001	3.04%	-5.34%	-3.62%	4.61%	0.86%	-0.80%	-0.11%	-3.09%	-4.77%	2.19%	3.77%	0.83%	-3.02%	1.55%	\$66,986	\$366,913
2002	-0.45%	-0.84%	1.99%	-2.30%	-0.40%	-3.97%	-4.22%	0.95%	-5.44%	4.26%	3.58%	-2.59%	-9.52%	2.38%	\$68,049	\$397,576
2003	-1.47%	-0.39%	0.58%	5.22%	4.34%	0.88%	0.06%	1.66%	0.41%	3.21%	0.96%	3.03%	19.87%	1.88%	\$69,593	\$553,834
2004	1.70%	1.29%	-0.37%	-2.31%	0.66%	1.47%	-1.89%	0.99%	1.15%	1.30%	2.52%	2.57%	9.32%	3.26%	\$71,456	\$681,395
2005	-1.34%	0.99%	-1.29%	-0.84%	2.76%	0.72%	2.11%	-0.10%	0.11%	-1.47%	2.58%	0.45%	4.65%	3.42%	\$73,881	\$789,481
2006	2.12%	0.15%	0.69%	0.54%	-1.96%	0.16%	0.50%	2.00%	1.68%	2.43%	1.80%	0.47%	11.02%	2.54%	\$76,264	\$957,438
2007	1.08%	-0.32%	0.65%	2.60%	1.90%	-1.15%	-1.72%	1.43%	2.45%	1.47%	-1.94%	-0.32%	6.16%	4.08%	\$78,440	\$1,096,157
2008	-2.91%	-1.78%	-0.19%	2.83%	0.98%	-4.98%	-0.44%	1.23%	-6.04%	-11.59%	-3.43%	2.44%	-22.21%	0.09%	\$81,451	\$923,184
2009	-5.18%	-6.48%	6.10%	6.40%	3.58%	0.50%	5.28%	2.62%	2.98%	-1.33%	3.88%	1.03%	20.05%	2.72%	\$81,162	\$1,200,617
2010	-1.50%	2.10%	3.74%	1.74%	-4.45%	-2.75%	4.50%	-2.33%	5.59%	2.52%	0.14%	3.66%	13.13%	1.50%	\$82,493	\$1,447,929
2011	1.36%	2.26%	0.23%	2.35%	-0.18%	-1.21%	-0.72%	-2.81%	-4.39%	6.98%	-0.28%	0.92%	4.14%	2.96%	\$85,097	\$1,593,687
2012	3.40%	2.49%	1.62%	0.04%	-3.34%	2.28%	1.17%	1.55%	1.58%	-1.01%	0.51%	0.66%	11.33%	1.74%	\$86,858	\$1,863,859
2013	3.03%	0.98%	2.36%	1.39%	0.74%	-1.41%	3.33%	-1.95%	2.56%	2.83%	1.60%	1.29%	17.91%	1.50%	\$88,130	\$2,292,535
2014	-1.24%	3.02%	0.21%	0.36%	1.71%	1.57%	-1.28%	2.95%	-1.57%	2.09%	1.74%	0.01%	9.84%	0.76%	\$89,560	\$2,612,004
2015	-0.77%	2.95%	-0.44%	0.10%	0.63%	-1.42%	1.31%	-3.72%	-1.45%	4.61%	0.24%	-1.41%	0.37%	0.73%	\$89,666	\$2,710,899
2016	-2.81%	0.28%	4.59%	0.54%	1.04%	0.93%	2.64%	0.06%	0.08%	-1.68%	1.61%	1.21%	8.63%	2.07%	\$90,797	\$3,039,656
2017	1.25%	2.48%	0.02%	0.93%	0.89%	0.55%	1.28%	0.42%	1.26%	1.31%	1.79%	0.78%	13.75%	2.11%	\$92,731	\$3,556,082
2018	2.71%	-2.61%	-0.93%	-0.09%	1.93%	0.40%	2.01%	2.31%	-0.13%	-4.76%	1.48%	-4.93%	-2.97%	1.91%	\$94,996	\$3,541,525
2019	5.55%	2.10%	1.65%	2.39%	-3.21%	4.63%	0.97%	-0.11%	0.80%	1.39%	2.24%	1.64%	21.67%	2.29%	\$96,718	\$4,412,965
2020	0.82%	-4.22%	-8.48%	8.55%	3.45%	1.73%	3.99%	3.86%	-2.12%	-1.48%	7.66%	2.70%	16.26%	1.36%	\$97,911	\$5,241,476
2021	-0.50%	1.27%	1.56%	3.44%	0.36%	1.87%	1.47%	1.66%	-3.12%	4.04%	-0.77%	2.17%	14.09%	7.04%	\$102,511	\$6,088,724
2022	-4.51%	-1.99%	0.84%	-6.99%	0.12%	-5.67%	6.50%	-3.32%	-7.27%	4.27%	4.60%	-3.83%	-16.97%	6.45%	\$110,714	\$5,161,082
2023	5.40%	-2.42%	2.61%	0.87%	-0.22%	3.93%	2.15%	-1.45%	-3.89%	-2.21%	7.40%	4.69%	17.44%	3.35%	\$115,272	\$6,186,256
2024	0.61%	2.72%	2.29%	-3.66%	3.53%	2.27%	2.00%	1.84%	1.78%	-1.40%	4.43%	-2.50%	14.46%	2.89%	\$118,672	\$7,205,914
2025	2.08%	-0.34%	-3.54%	-0.06%	3.58%	3.70%	1.30%	1.84%	2.52%	1.56%	0.41%	-0.14%	13.46%	2.68%	\$121,884	\$8,306,334
2026	1.04%	0.27%	-3.69%	6.24%	3.27%	-0.12%							6.92%	3.05%	\$62,558	\$8,945,861

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Drawdowns for Historical Market Stress Periods

Stress Period	Start	End	Dual Momentum Model	Vanguard Balanced Index Inv
Asian Crisis	Jul 1997	Jan 1998	-5.61%	-2.61%
Russian Debt Default	Jul 1998	Oct 1998	-15.38%	-9.66%
Dotcom Crash	Mar 2000	Oct 2002	-8.46%	-22.39%
Subprime Crisis	Nov 2007	Mar 2009	-13.56%	-32.57%
COVID-19 Start	Jan 2020	Mar 2020	-8.29%	-12.34%

Drawdowns for Dual Momentum Model (worst 10)

Rank	Start	End	Length	Recovery By	Recovery Time	Underwater Period	Drawdown
1	Jan 2022	Oct 2022	10 months	Mar 2024	1 year 5 months	2 years 3 months	-21.66%
2	Mar 2015	Jan 2016	11 months	Aug 2017	1 year 7 months	2 years 6 months	-15.72%
3	Jul 1998	Aug 1998	2 months	Jan 1999	5 months	7 months	-15.38%
4	Nov 2007	Jun 2008	8 months	May 2009	11 months	1 year 7 months	-13.56%
5	Mar 2026	Mar 2026	1 month				-8.64%
6	Nov 2001	Jun 2002	8 months	Feb 2003	8 months	1 year 4 months	-8.46%
7	Jan 2020	Feb 2020	2 months	Jul 2020	5 months	7 months	-8.29%
8	May 2010	Aug 2010	4 months	Sep 2010	1 month	5 months	-8.15%
9	Mar 2005	Aug 2005	6 months	Dec 2005	4 months	10 months	-6.93%
10	Oct 2018	Oct 2018	1 month	Mar 2019	5 months	6 months	-6.85%

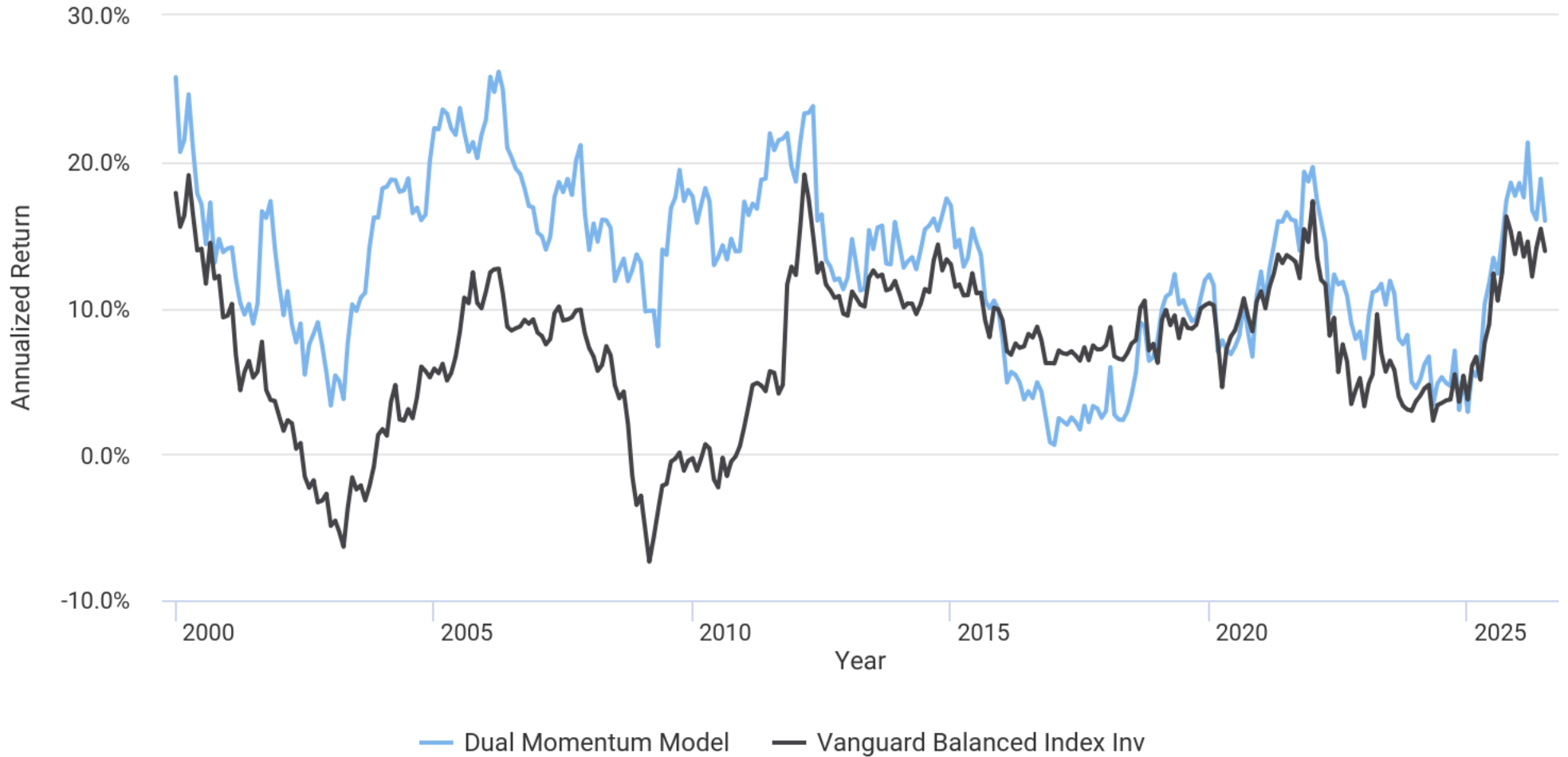
Drawdowns for Vanguard Balanced Index Inv (worst 10)

Rank	Start	End	Length	Recovery By	Recovery Time	Underwater Period	Drawdown
1	Nov 2007	Feb 2009	1 year 4 months	Dec 2010	1 year 10 months	3 years 2 months	-32.57%
2	Sep 2000	Sep 2002	2 years 1 month	Feb 2004	1 year 5 months	3 years 6 months	-22.39%
3	Jan 2022	Sep 2022	9 months	Feb 2024	1 year 5 months	2 years 2 months	-20.85%
4	Feb 2020	Mar 2020	2 months	Jun 2020	3 months	5 months	-12.34%
5	Jul 1998	Aug 1998	2 months	Nov 1998	3 months	5 months	-9.66%
6	May 2011	Sep 2011	5 months	Jan 2012	4 months	9 months	-9.03%
7	Sep 2018	Dec 2018	4 months	Mar 2019	3 months	7 months	-8.23%
8	Apr 2000	May 2000	2 months	Aug 2000	3 months	5 months	-5.35%
9	Jun 2015	Sep 2015	4 months	Apr 2016	7 months	11 months	-5.23%
10	Dec 2024	Apr 2025	5 months	Jun 2025	2 months	7 months	-4.38%

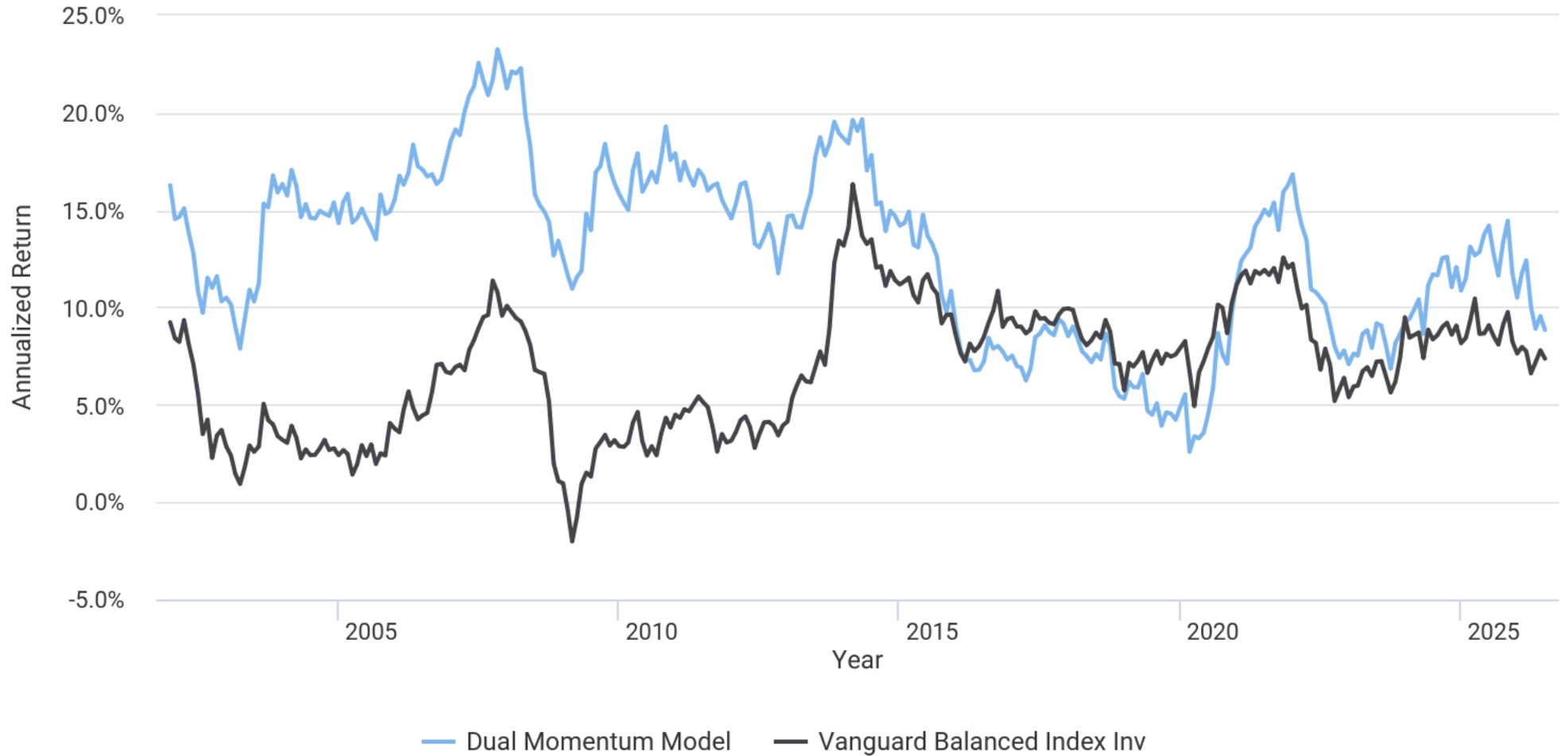
Rolling Returns (Jan 1997 - Jun 2026)

Roll Period	Dual Momentum Model			Vanguard Balanced Index Inv		
	Average	High	Low	Average	High	Low
1 year	14.01%	53.85%	-21.40%	8.12%	36.16%	-27.67%
3 years	12.90%	26.15%	0.62%	7.13%	19.12%	-7.37%
5 years	12.85%	23.26%	2.58%	6.90%	16.33%	-2.02%
7 years	12.98%	21.23%	6.43%	7.08%	12.28%	0.31%
10 years	12.84%	18.37%	7.02%	7.07%	11.55%	0.90%
15 years	12.95%	17.18%	9.51%	7.22%	10.55%	4.95%

Annualized Rolling Return (36 months)



Annualized Rolling Return (60 months)



Risk Management Performance

Type	Periods #	Periods %	Model						Benchmark			Risk-free Return	Up/Down Capture			
			Return	Stdev	SR	XS	TE	IR	Return	Stdev	SR		Win %	Upside	Downside	Gain/Loss
In Market	274	77.40%	14.95%	13.41%	0.94	5.63%	6.96%	0.81	9.32%	8.09%	0.87	2.27%	63.50%	165.58	144.01	1.08
Out of Market	80	22.60%	8.74%	5.89%	1.09	5.46%	15.70%	0.35	3.28%	14.25%	0.14	2.23%	53.75%	11.05	-39.06	1.05

Risk management regimes are based on monthly returns. All risk and return metrics are annualized. Gain/loss ratio for average periodic excess return vs. benchmark.

Trading Performance

Metric	Value
Total Trades	96
Analyzed Trades	95
Rebalancing Trades	0
Winning Trades	54
Losing Trades	41
Win %	56.84%
Average Win	4.81%
Median Win	1.89%
Average Loss	2.14%
Median Loss	2.06%
Average Annual Turnover	308.32%
Average Turnover Per Trade	100.00%
Median Turnover Per Trade	100.00%
Average Turnover Per Trade Excluding Rebalancing	100.00%
Median Turnover Per Trade Excluding Rebalancing	100.00%
Cumulative Trading Return without Costs	4,109.84%
Cumulative Trading Return after Costs	2,569.87%
Cumulative Trading Cost Impact	1,539.98%
Annualized Trading Return without Costs	13.52%
Annualized Trading Return after Costs	11.78%
Annualized Trading Cost Impact	1.74%

Trading costs estimate assumes 25bps cost per trade.

Model trades

#	Start Date	End Date	Trade Date	Trade Type	Turnover	Allocation	Dual Momentum Model	Vanguard Balanced Index Inv
1	01/01/1997	08/31/1998	12/31/1999	Initial Entry	N/A	100.00% Vanguard 500 Index Investor (VFINX)	32.62%	22.32%
2	09/01/1998	10/31/1998	08/31/1998	Signal	100.00%	100.00% Vanguard Interm-Term Treasury Inv (VFITX)	2.84%	8.97%
3	11/01/1998	07/31/1999	10/30/1998	Signal	100.00%	100.00% Vanguard 500 Index Investor (VFINX)	22.13%	12.76%
4	08/01/1999	03/31/2000	07/30/1999	Signal	100.00%	100.00% Vanguard Total Intl Stock Index Inv (VGTSX)	19.20%	12.49%
5	04/01/2000	06/30/2000	03/31/2000	Signal	100.00%	100.00% Vanguard 500 Index Investor (VFINX)	-2.61%	-2.07%
6	07/01/2000	08/31/2000	06/30/2000	Signal	100.00%	100.00% Vanguard Interm-Term Treasury Inv (VFITX)	2.25%	4.07%
7	09/01/2000	09/30/2000	08/31/2000	Signal	100.00%	100.00% Vanguard 500 Index Investor (VFINX)	-5.29%	-2.53%
8	10/01/2000	12/31/2000	09/29/2000	Signal	100.00%	100.00% Vanguard Interm-Term Treasury Inv (VFITX)	13.50%	-7.42%
9	01/01/2001	01/31/2001	12/31/2000	Signal	100.00%	100.00% Vanguard 500 Index Investor (VFINX)	-1.47%	-0.45%
10	02/01/2001	03/31/2001	01/31/2001	Signal	100.00%	100.00% Vanguard Interm-Term Treasury Inv (VFITX)	-1.04%	1.13%
11	04/01/2001	06/30/2001	03/28/2001	Signal	100.00%	100.00% Vanguard Total Intl Stock Index Inv (VGTSX)	-2.75%	-6.55%
12	07/01/2001	03/31/2002	06/28/2001	Signal	100.00%	100.00% Vanguard Interm-Term Treasury Inv (VFITX)	10.05%	-5.06%
13	04/01/2002	06/30/2002	03/31/2002	Signal	100.00%	100.00% Vanguard 500 Index Investor (VFINX)	15.39%	10.75%
14	07/01/2002	06/30/2002	06/30/2002	Signal	100.00%	100.00% Vanguard Total Intl Stock Index Inv (VGTSX)	32.85%	12.29%
15	07/01/2002	07/31/2002	06/30/2002	Signal	100.00%	100.00% Vanguard 500 Index Investor (VFINX)	-3.31%	-1.89%
16	08/01/2002	09/30/2002	07/30/2002	Signal	100.00%	100.00% Vanguard Interm-Term Treasury Inv (VFITX)	2.27%	2.15%
17	10/01/2002	05/31/2003	09/30/2002	Signal	100.00%	100.00% Vanguard Total Intl Stock Index Inv (VGTSX)	13.24%	6.75%
18	06/01/2003	06/30/2003	05/31/2003	Signal	100.00%	100.00% Vanguard 500 Index Investor (VFINX)	0.13%	0.72%
19	07/01/2003	07/31/2003	06/30/2003	Signal	100.00%	100.00% Vanguard Interm-Term Treasury Inv (VFITX)	-1.41%	2.11%

#	Start Date	End Date	Trade Date	Trade Type	Turnover	Allocation	Dual Momentum Model	Vanguard Balanced Index Inv
20	08/01/2005	08/31/2005	07/29/2005	Signal	100.00%	100.00% Vanguard 500 Index Investor (VFINX)	-0.91%	-0.10%
21	09/01/2005	07/31/2006	08/31/2005	Signal	100.00%	100.00% Vanguard Total Intl Stock Index Inv (VGTSX)	21.03%	3.84%
22	08/01/2006	08/31/2006	07/31/2006	Signal	100.00%	100.00% Vanguard Interm-Term Treasury Inv (VFITX)	1.62%	2.00%
23	09/01/2006	09/30/2006	08/31/2006	Signal	100.00%	100.00% Vanguard Total Intl Stock Index Inv (VGTSX)	0.25%	1.68%
24	10/01/2006	11/30/2006	09/29/2006	Signal	100.00%	100.00% Vanguard 500 Index Investor (VFINX)	5.20%	4.28%
25	12/01/2006	12/31/2007	11/30/2006	Signal	100.00%	100.00% Vanguard Total Intl Stock Index Inv (VGTSX)	19.09%	6.66%
26	01/01/2008	05/31/2008	12/31/2007	Signal	100.00%	100.00% Vanguard Interm-Term Treasury Inv (VFITX)	1.84%	-1.16%
27	06/01/2008	06/30/2008	05/30/2008	Signal	100.00%	100.00% Vanguard Total Intl Stock Index Inv (VGTSX)	-8.99%	-4.98%
28	07/01/2008	04/30/2009	06/30/2008	Signal	100.00%	100.00% Vanguard Interm-Term Treasury Inv (VFITX)	8.65%	-17.09%
29	05/01/2009	11/30/2009	04/30/2009	Signal	100.00%	100.00% Vanguard Total Intl Stock Index Inv (VGTSX)	37.20%	18.71%
30	12/01/2009	05/31/2010	11/30/2009	Signal	100.00%	100.00% Vanguard 500 Index Investor (VFINX)	0.36%	2.47%
31	06/01/2010	07/31/2010	05/28/2010	Signal	100.00%	100.00% Vanguard Interm-Term Treasury Inv (VFITX)	3.16%	1.63%
32	08/01/2010	11/30/2010	07/30/2010	Signal	100.00%	100.00% Vanguard Total Intl Stock Index Inv (VGTSX)	5.65%	5.89%
33	12/01/2010	12/31/2010	11/30/2010	Signal	100.00%	100.00% Vanguard 500 Index Investor (VFINX)	6.67%	3.66%
34	01/01/2011	01/31/2011	12/31/2010	Signal	100.00%	100.00% Vanguard Total Intl Stock Index Inv (VGTSX)	0.70%	1.36%
35	02/01/2011	07/31/2011	01/31/2011	Signal	100.00%	100.00% Vanguard 500 Index Investor (VFINX)	1.39%	2.71%
36	08/01/2011	10/31/2011	07/29/2011	Signal	100.00%	100.00% Vanguard Interm-Term Treasury Inv (VFITX)	3.02%	-0.60%
37	11/01/2011	11/30/2011	10/31/2011	Signal	100.00%	100.00% Vanguard 500 Index Investor (VFINX)	-0.23%	-0.28%
38	12/01/2011	12/31/2011	11/30/2011	Signal	100.00%	100.00% Vanguard Interm-Term Treasury Inv (VFITX)	1.00%	0.92%
39	01/01/2012	05/31/2012	12/30/2011	Signal	100.00%	100.00% Vanguard 500 Index Investor (VFINX)	5.09%	4.13%

#	Start Date	End Date	Trade Date	Trade Type	Turnover	Allocation	Dual Momentum Model	Vanguard Balanced Index Inv
40	06/01/2012	06/30/2012	05/31/2012	Signal	100.00%	100.00% Vanguard Interm-Term Treasury Inv (VFITX)	-0.22%	2.28%
41	07/01/2012	10/31/2012	06/29/2012	Signal	100.00%	100.00% Vanguard 500 Index Investor (VFINX)	4.33%	3.32%
42	11/01/2012	03/31/2013	10/31/2012	Signal	100.00%	100.00% Vanguard Total Intl Stock Index Inv (VGTSX)	9.09%	7.75%
43	04/01/2013	09/30/2013	03/28/2013	Signal	100.00%	100.00% Vanguard 500 Index Investor (VFINX)	8.22%	4.63%
44	10/01/2013	10/31/2013	09/30/2013	Signal	100.00%	100.00% Vanguard Total Intl Stock Index Inv (VGTSX)	3.41%	2.83%
45	11/01/2013	04/30/2015	10/31/2013	Signal	100.00%	100.00% Vanguard 500 Index Investor (VFINX)	22.13%	15.08%
46	05/01/2015	07/31/2015	04/30/2015	Signal	100.00%	100.00% Vanguard Total Intl Stock Index Inv (VGTSX)	-4.38%	0.50%
47	08/01/2015	08/31/2015	07/31/2015	Signal	100.00%	100.00% Vanguard 500 Index Investor (VFINX)	-6.05%	-3.72%
48	09/01/2015	10/31/2015	08/31/2015	Signal	100.00%	100.00% Vanguard Interm-Term Treasury Inv (VFITX)	0.71%	3.09%
49	11/01/2015	01/31/2016	10/30/2015	Signal	100.00%	100.00% Vanguard 500 Index Investor (VFINX)	-6.22%	-3.95%
50	02/01/2016	03/31/2016	01/29/2016	Signal	100.00%	100.00% Vanguard Interm-Term Treasury Inv (VFITX)	1.04%	4.88%
51	04/01/2016	04/30/2016	03/31/2016	Signal	100.00%	100.00% Vanguard 500 Index Investor (VFINX)	0.37%	0.54%
52	05/01/2016	05/31/2016	04/29/2016	Signal	100.00%	100.00% Vanguard Total Intl Stock Index Inv (VGTSX)	-1.02%	1.04%
53	06/01/2016	08/31/2016	05/31/2016	Signal	100.00%	100.00% Vanguard 500 Index Investor (VFINX)	4.07%	3.67%
54	09/01/2016	11/30/2016	08/31/2016	Signal	100.00%	100.00% Vanguard Total Intl Stock Index Inv (VGTSX)	-2.52%	-0.02%
55	12/01/2016	03/31/2017	11/30/2016	Signal	100.00%	100.00% Vanguard 500 Index Investor (VFINX)	8.11%	5.04%
56	04/01/2017	04/30/2017	03/31/2017	Signal	100.00%	100.00% Vanguard Total Intl Stock Index Inv (VGTSX)	2.20%	0.93%
57	05/01/2017	05/31/2017	04/28/2017	Signal	100.00%	100.00% Vanguard 500 Index Investor (VFINX)	1.39%	0.89%
58	06/01/2017	11/30/2017	05/31/2017	Signal	100.00%	100.00% Vanguard Total Intl Stock Index Inv (VGTSX)	9.43%	6.80%
59	12/01/2017	03/31/2018	11/30/2017	Signal	100.00%	100.00% Vanguard 500 Index Investor (VFINX)	0.30%	-0.13%

#	Start Date	End Date	Trade Date	Trade Type	Turnover	Allocation	Dual Momentum Model	Vanguard Balanced Index Inv
60	04/01/2018	04/30/2018	03/29/2018	Signal	100.00%	100.00% Vanguard Total Intl Stock Index Inv (VGTSX)	0.83%	-0.09%
61	05/01/2018	05/31/2018	04/30/2018	Signal	100.00%	100.00% Vanguard Interm-Term Treasury Inv (VFITX)	0.76%	1.93%
62	06/01/2018	10/31/2018	05/31/2018	Signal	100.00%	100.00% Vanguard 500 Index Investor (VFINX)	0.90%	-0.34%
63	11/01/2018	01/31/2019	10/31/2018	Signal	100.00%	100.00% Vanguard Interm-Term Treasury Inv (VFITX)	3.48%	1.83%
64	02/01/2019	02/28/2019	01/31/2019	Signal	100.00%	100.00% Vanguard 500 Index Investor (VFINX)	3.20%	2.10%
65	03/01/2019	03/31/2019	02/28/2019	Signal	100.00%	100.00% Vanguard Total Intl Stock Index Inv (VGTSX)	0.72%	1.65%
66	04/01/2019	05/31/2019	03/29/2019	Signal	100.00%	100.00% Vanguard 500 Index Investor (VFINX)	-2.58%	-0.89%
67	06/01/2019	06/30/2019	05/31/2019	Signal	100.00%	100.00% Vanguard Interm-Term Treasury Inv (VFITX)	1.02%	4.63%
68	07/01/2019	10/31/2019	06/28/2019	Signal	100.00%	100.00% Vanguard 500 Index Investor (VFINX)	3.86%	3.08%
69	11/01/2019	11/30/2019	10/31/2019	Signal	100.00%	100.00% Vanguard Total Intl Stock Index Inv (VGTSX)	1.11%	2.24%
70	12/01/2019	02/29/2020	11/29/2019	Signal	100.00%	100.00% Vanguard 500 Index Investor (VFINX)	-5.53%	-1.85%
71	03/01/2020	05/31/2020	02/28/2020	Signal	100.00%	100.00% Vanguard Interm-Term Treasury Inv (VFITX)	3.18%	2.77%
72	06/01/2020	10/31/2020	05/29/2020	Signal	100.00%	100.00% Vanguard 500 Index Investor (VFINX)	8.08%	5.95%
73	11/01/2020	03/31/2021	10/30/2020	Signal	100.00%	100.00% Vanguard Total Intl Stock Index Inv (VGTSX)	24.22%	13.16%
74	04/01/2021	01/31/2022	03/31/2021	Signal	100.00%	100.00% Vanguard 500 Index Investor (VFINX)	14.81%	6.45%
75	02/01/2022	03/31/2022	01/31/2022	Signal	100.00%	100.00% Vanguard Interm-Term Treasury Inv (VFITX)	-3.39%	-1.16%
76	04/01/2022	04/30/2022	03/31/2022	Signal	100.00%	100.00% Vanguard 500 Index Investor (VFINX)	-8.73%	-6.99%
77	05/01/2022	07/31/2022	04/29/2022	Signal	100.00%	100.00% Vanguard Interm-Term Treasury Inv (VFITX)	1.83%	0.58%
78	08/01/2022	08/31/2022	07/29/2022	Signal	100.00%	100.00% Vanguard 500 Index Investor (VFINX)	-4.09%	-3.32%
79	09/01/2022	11/30/2022	08/31/2022	Signal	100.00%	100.00% Vanguard Interm-Term Treasury Inv (VFITX)	-1.61%	1.14%

#	Start Date	End Date	Trade Date	Trade Type	Turnover	Allocation	Dual Momentum Model	Vanguard Balanced Index Inv
80 2	12/01/202	05/31/202	11/30/202	Signal	100.00%	100.00% Vanguard Total Intl Stock Index Inv (VGTSX)	2.52%	2.16%
81 3	06/01/202	09/30/202	05/31/202	Signal	100.00%	100.00% Vanguard 500 Index Investor (VFINX)	3.07%	0.56%
82 3	10/01/202	11/30/202	09/29/202	Signal	100.00%	100.00% Vanguard Interm-Term Treasury Inv (VFITX)	2.21%	5.03%
83 3	12/01/202	09/30/202	11/30/202	Signal	100.00%	100.00% Vanguard 500 Index Investor (VFINX)	27.47%	19.36%
84 4	10/01/202	10/31/202	09/30/202	Signal	100.00%	100.00% Vanguard Total Intl Stock Index Inv (VGTSX)	-4.71%	-1.40%
85 4	11/01/202	02/28/202	10/31/202	Signal	100.00%	100.00% Vanguard 500 Index Investor (VFINX)	4.79%	3.59%
86 5	03/01/202	03/31/202	02/28/202	Signal	100.00%	100.00% Vanguard Total Intl Stock Index Inv (VGTSX)	0.27%	-3.54%
87 5	04/01/202	04/30/202	03/31/202	Signal	100.00%	100.00% Vanguard Interm-Term Treasury Inv (VFITX)	1.23%	-0.06%
88 5	05/01/202	07/31/202	04/30/202	Signal	100.00%	100.00% Vanguard Total Intl Stock Index Inv (VGTSX)	7.66%	8.81%
89 5	08/01/202	08/31/202	07/31/202	Signal	100.00%	100.00% Vanguard 500 Index Investor (VFINX)	2.02%	1.84%
90 5	09/01/202	09/30/202	08/29/202	Signal	100.00%	100.00% Vanguard Total Intl Stock Index Inv (VGTSX)	3.62%	2.52%
91 5	10/01/202	12/31/202	09/30/202	Signal	100.00%	100.00% Vanguard 500 Index Investor (VFINX)	2.62%	1.84%
92 6	01/01/202	03/31/202	12/31/202	Signal	100.00%	100.00% Vanguard Total Intl Stock Index Inv (VGTSX)	1.73%	-2.43%
93 6	04/01/202	04/30/202	03/31/202	Signal	100.00%	100.00% Vanguard Interm-Term Treasury Inv (VFITX)	0.02%	6.24%
94 6	05/01/202	05/31/202	04/30/202	Signal	100.00%	100.00% Vanguard Total Intl Stock Index Inv (VGTSX)	3.75%	3.27%
95 6	06/01/202	06/30/202	05/29/202	Signal	100.00%	100.00% Vanguard 500 Index Investor (VFINX)	-0.96%	-0.12%
96 6	07/01/202	07/31/202	06/30/202	Open Trade	100.00%	100.00% Vanguard Total Intl Stock Index Inv (VGTSX)	N/A	N/A

Trade execution assumes end-of-day close price minus trade execution cost. Initial entry is excluded from turnover and transaction costs.

Notes:

- **IMPORTANT:** The projections or other information generated by Portfolio Visualizer regarding the likelihood of various investment outcomes are hypothetical in nature, do not reflect actual investment results and are not guarantees of future results. Results may vary with each use and over time.
- The results do not constitute investment advice or recommendation, are provided solely for informational purposes, and are not an offer to buy or sell any securities. All use is subject to terms of service.
- Investing involves risk, including possible loss of principal. Past performance is not a guarantee of future results.
- Asset allocation and diversification strategies do not guarantee a profit or protect against a loss.
- Hypothetical returns do not reflect trading costs, transaction fees, commissions, or actual taxes due on investment returns.
- The results are based on information from a variety of sources we consider reliable, but we do not represent that the information is accurate or complete.
- Refer to the related documentation sections for more details on terms and definitions, methodology, and data sources.
- The simulated performance results are created by retroactively applying the specified investment strategy to historical data. Future results may vary.
- The results are based on the total return of assets and assume that all received dividends and distributions are reinvested.
- Compound annualized growth rate (CAGR) is the annualized geometric mean return of the portfolio. It is calculated from the portfolio start and end balance and is thus impacted by any cashflows.
- The time-weighted rate of return (TWRR) is a measure of the compound rate of growth in a portfolio. This is calculated from the holding period returns (e.g. monthly returns), and TWRR will thus not be impacted by cashflows. If there are no external cashflows, TWRR will equal CAGR.
- The money-weighted rate of return (MWRR) is the internal rate of return (IRR) taking into account cashflows. This is the discount rate at which the present value of cash inflows equals the present value of cash outflows.
- Total return is the combined return in income and capital appreciation from investment in an asset. Yield measures the current cash income received from investment in an asset. Bonds provide yield in the form of interest payments and stocks through dividends.
- Standard deviation (Stddev) is used to measure the dispersion of returns around the mean and is often used as a measure of risk. A higher standard deviation implies greater the dispersion of data points around the mean.
- Sharpe Ratio is a measure of risk-adjusted performance of the portfolio, and it is calculated by dividing the mean monthly excess return of the portfolio over the risk-free rate by the standard deviation of excess return, and the displayed value is annualized.
- Sortino Ratio is a measure of risk-adjusted return which is a modification of the Sharpe Ratio. While the latter is the ratio of average returns in excess of a risk-free rate divided by the standard deviation of those excess returns, the Sortino Ratio has the same denominator divided by the standard deviation of returns below the risk-free rate.
- Treynor Ratio is a measure of risk-adjusted performance of the portfolio. It is similar to the Sharpe Ratio, but it uses portfolio beta (systematic risk) as the risk metric in the denominator.
- Calmar Ratio is a measure of risk-adjusted performance of the portfolio. It is calculated as the annualized return over the past 36 months divided by the maximum drawdown over the past 36 months based on monthly returns.
- Downside deviation measures the downside volatility of the portfolio returns unlike standard deviation, which includes both upside and downside deviations. Downside deviation is calculated based on negative returns that hurt the portfolio performance.
- Risk-free returns are calculated based on U.S. 3-Month Treasury Bill Rate.
- Inflation is calculated based on U.S. Consumer Price Index.
- Skewness is a measure of the asymmetry of the probability distribution or returns from a normal Gaussian distribution shape about its mean. Negative skewness is associated with the left (typically negative returns) tail of the distribution extending further than the right tail; and positive skewness is associated with the right (typically positive returns) tail of the distribution extending further than the left tail.
- Excess kurtosis is a measure of whether a data distribution is peaked or flat relative to a normal distribution. Distributions with high kurtosis tend to have a distinct peak near the mean, decline rather rapidly, and have heavy or fat tails.
- A drawdown refers to the decline in value of a single investment or an investment portfolio from a relative peak value to a relative trough. A maximum drawdown (Max Drawdown) is the maximum observed loss from a peak to a trough of a portfolio before a new peak is attained. Drawdown values are calculated based on monthly returns.
- Value at Risk (VaR) measures the scale of loss at a given confidence level. For example, if the 95% confidence one-month VaR is 3%, there is 95% confidence that over the next month the portfolio will not lose more than 3%. Value at Risk can be calculated directly based on historical returns based on a given percentile or analytically based on the mean and standard deviation of the returns.
- Conditional Value at Risk (CVaR) measures the scale of the expected loss once the specific Value at Risk (VaR) breakpoint has been breached, i.e., it calculates the average tail loss by taking a weighted average between the value at risk and losses exceeding the value at risk.
- Beta is a measure of systematic risk and measures the volatility of a particular investment relative to the market or its benchmark. Alpha measures the active return of the investment compared to the market benchmark return. R-squared is the percentage of a portfolio's movements that can be explained by movements in the selected benchmark index.
- Active return is the investment return minus the return of its benchmark. For periods longer than 12 months this is displayed as annualized value, i.e., annualized investment return minus annualized benchmark return.
- Tracking error, also known as active risk, is the standard deviation of active return. This is displayed as annualized value based on the standard deviation of monthly active returns.
- Information ratio is the active return divided by the tracking error. It measures whether the investment outperformed its benchmark consistently.
- Gain/Loss ratio is a measure of downside risk, and it is calculated as the average positive return in up periods divided by the average negative return in down periods.
- Upside Capture Ratio measures how well the fund performed relative to the benchmark when the market was up, and Downside Capture Ratio measures how well the fund performed relative to the benchmark when the market was down. An upside capture ratio greater than 100 would indicate that the fund outperformed its benchmark when the market was up, and a downside capture ratio below 100 would indicate that the fund lost less than its benchmark when the market was down. To calculate upside capture ratio a new series from the portfolio returns is constructed by dropping all time periods where the benchmark return is less than equal to zero. The up capture is then the quotient of the annualized return of the resulting manager series, divided by the annualized return of the resulting benchmark series. The downside capture ratio is calculated analogously.
- All risk measures for the portfolio and portfolio assets are calculated based on monthly returns.
- The annual results for 2026 are based on monthly returns from January to June.
- The selected portfolio allocation for comparison assumes monthly rebalancing.

- An equal-weight allocation is used for the assets held by the momentum model.
- The results for the tactical asset allocation model assume monthly rebalancing trades.
- The relative strength signal is based on total return including dividends.
- Inflation adjusted monthly contribution of \$5,000 was applied at the end of each period. This is reflected in the CAGR and maximum drawdown shown above.